

VoxSense: Emotion-Aware Speech Interaction Assistant

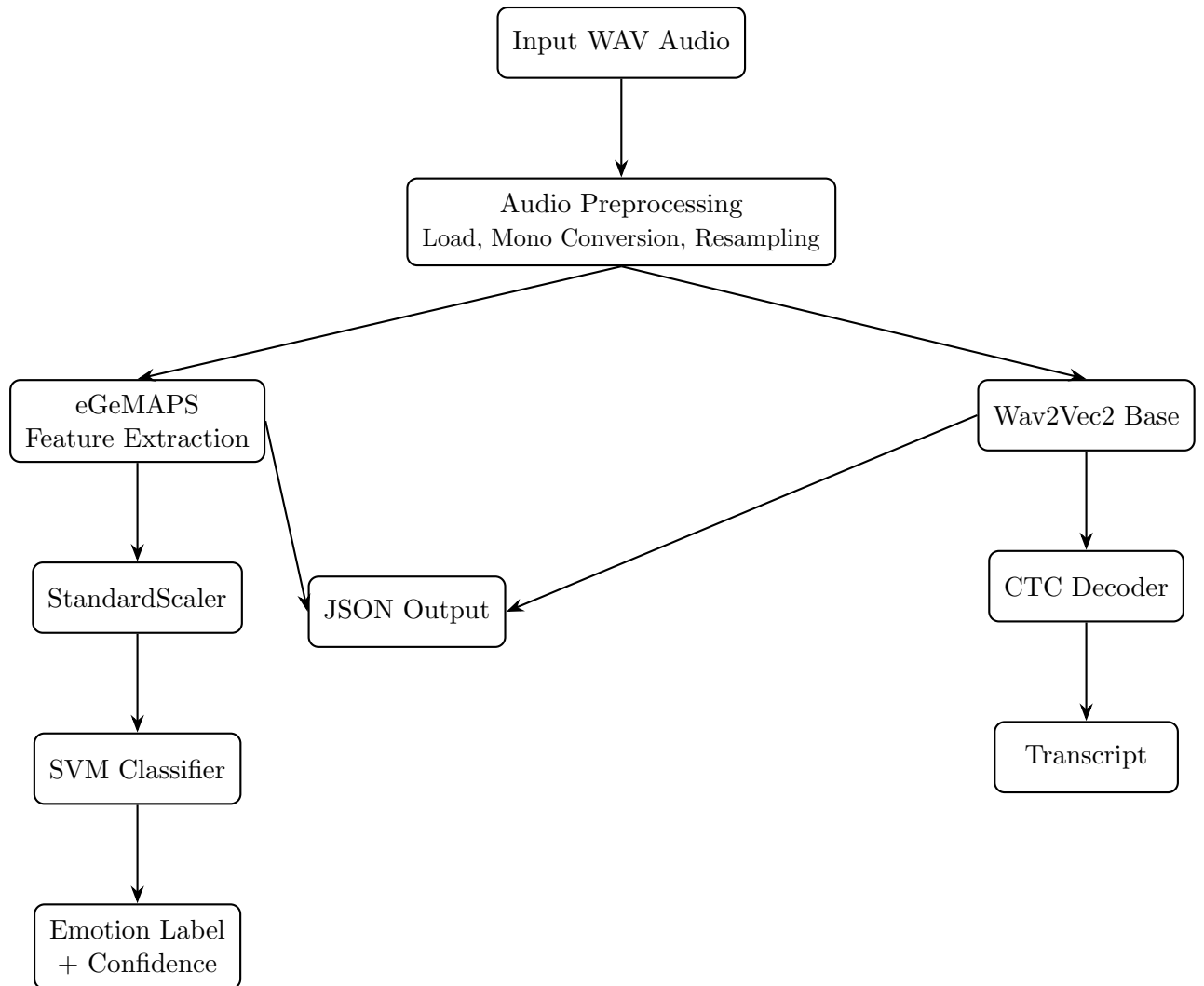
AudialMind Seasons of Code 2026 Final Project

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1 Introduction

VoxSense is an end-to-end speech processing system that combines Speech Emotion Recognition (SER) with Automatic Speech Recognition (ASR). It takes a single speech recording as input, predicts the emotion of the speaker with the help of an eGeMAPS+SVM pipeline, creates a transcript of the speech using Wav2Vec2, and outputs these together as a JSON response.

2 System Architecture



3 Methodology

3.1 Preprocessing

Librosa was used to load audio, ensure they are mono if needed and downsample them to the intended sampling rate. Waveform, STFT, and Mel spectrogram were visually inspected to confirm preprocessing was correct.

3.2 Speech Emotion Recognition

Emotion classification was conducted on RAVDESS. MFCC and eGeMAPS features were extracted and scaled using StandardScaler. The default model for SER was an RBF-kernel SVM with a 10-fold stratified cross-validation strategy. Transformer representations from Wav2Vec2 was additionally probed for effectiveness.

3.3 Automatic Speech Recognition

Transcribing was done using a finetuned facebook/wav2vec2-base-960h and measured with WER and CER on the LibriSpeech.

4 Experimental Results

Table 1: Performance Summary

Metric	Value
MFCC + SVM UAR	0.5410
eGeMAPS + SVM UAR	0.5820
Wav2Vec2 + LinearSVC UAR	0.5684
Best Transformer Layer	Layer 8 (0.6999)
WER	0.0337
CER	0.0337
Inference Time	≈ 0.38 s

Figures referenced in the report:

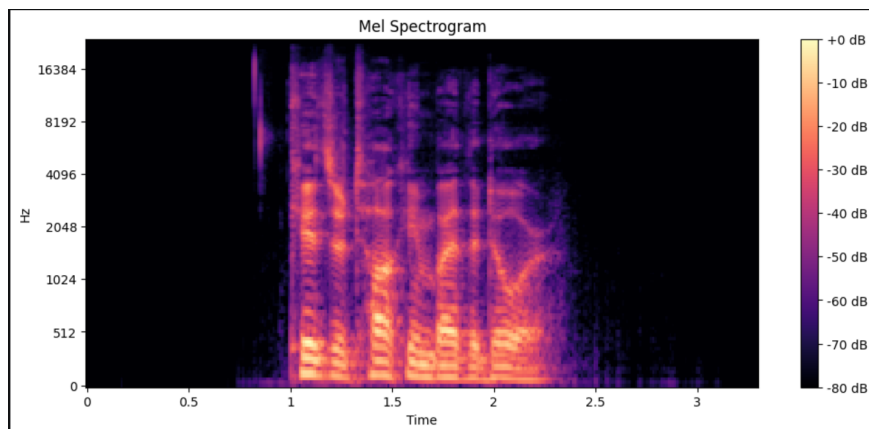


Figure 1: Mel spectrogram used during preprocessing.

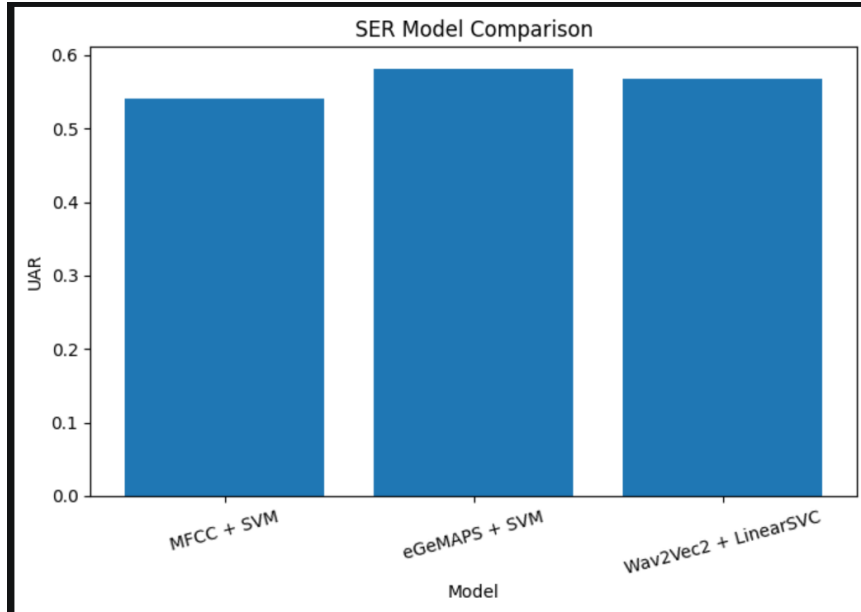


Figure 2: Comparison of SER models.

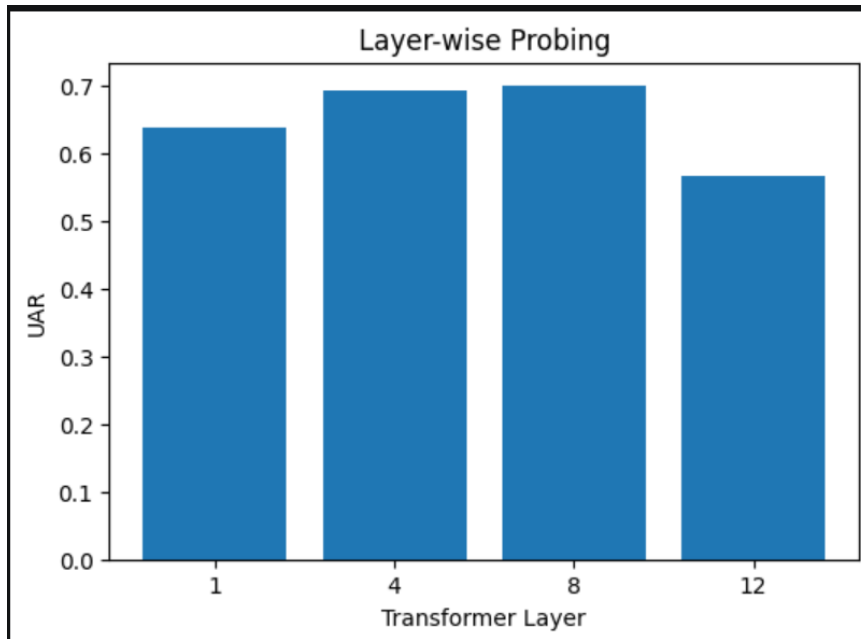


Figure 3: Layer-wise probing using frozen Wav2Vec2 embeddings.

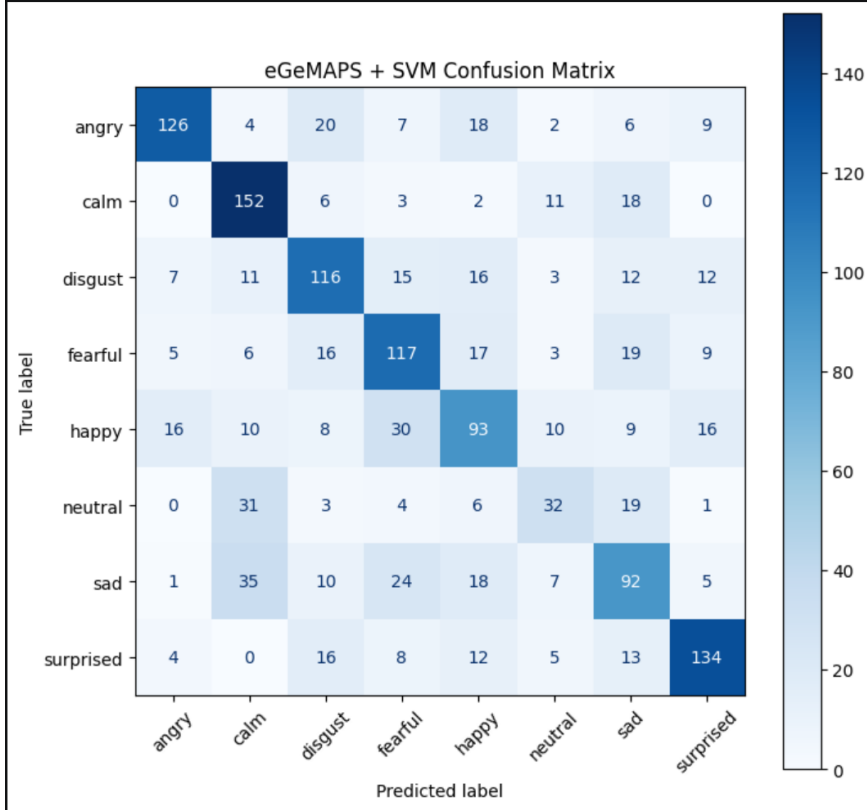


Figure 4: Confusion matrix of the final eGeMAPS+SVM model.

5 Discussion

The eGeMAPS representation outperformed the MFCC baseline while remaining computationally efficient. Layer probing indicated that intermediate transformer layers contained stronger emotion-discriminative representations than the final layer. The ASR module achieved low WER and CER, making it suitable for integration with the SER pipeline.

6 What I Would Do With More Compute and Time

With more development time and computational resources, there are a few directions to explore to make the proposed system even more efficient and robust: Firstly, rather than using the frozen Wav2Vec2 embeddings to perform speech emotion recognition, I would attempt to fine-tune the entire transformer model on the RAVDESS dataset. Fine-tuning would help the model to capture the characteristics of different emotions, rather than just generic speech features. Secondly, I would perform experiments on larger, and more diverse emotion recognition datasets such as the IEMOCAP and MELD datasets.

Such datasets would provide larger variation in the speaker and recording environment, and more conversational speech for improved generalisation.

For the Automatic Speech Recognition part, I would use beam-search decoding combined with an external language model to lower the substitution and insertion errors experienced during evaluation, especially in long sentences and in the case of acoustically similar words. Lastly, I would turn this prototype into a real-time application that takes audio input from a microphone and streams audio on the fly. This involves low-latency inference, real-time audio stream handling and the construction of a real-time user interface for conversational and assistance purposes.

7 Failure Analysis

Although the proposed system achieved low Word Error Rate (WER) and Character Error Rate (CER), certain transcription errors were observed during qualitative evaluation. Most errors occurred when words had similar acoustic patterns or when speech was spoken rapidly. Examples of common errors are shown in Table 2.

Table 2: Representative ASR Failure Cases

Ground Truth	Prediction
FLOUR FATTENED SAUCE STUFF IT INTO	FLOWER FAT AND SAUCE STUFFERED INTO
HELLO BERTIE ANY GOOD IN YOUR MIND	HO BERTIE ANY GOOD IN YOUR MIND

These examples indicate that the ASR model occasionally confuses phonetically similar words, performs word insertions or deletions, and struggles with rapidly spoken phrases. Such errors are common in end-to-end speech recognition systems and can become more pronounced in noisy environments or for speakers with varying accents. The SER model also exhibited occasional confusion between acoustically similar emotions such as *calm* and *neutral*, which is reflected in the confusion matrix presented earlier.

8 Conclusion

VoxSense successfully integrates speech emotion recognition and automatic speech recognition into a single application. The project demonstrates the effectiveness of combining classical acoustic features with modern transformer-based speech models for practical speech analysis.